

**ON-DEVICE ENCRYPTION, KEY MANAGEMENT, SECURE COMMUNICATIONS, AND
PRIVACY-PRESERVING OUTPUTS (NON-LIMITING)**

A. Technical Field

This section relates to data security for physiologic sensing devices and associated computing systems, including on-device encryption, secure storage, secure communications, and privacy-preserving presentation of derived metrics.

B. Summary (Non-Limiting)

In various embodiments, one or more nodes described in this application store and/or transmit time-series sensor data and derived metrics. To protect user privacy and reduce risk of data exposure, a security architecture may be implemented including on-device encryption for data at rest, secure key storage, authenticated encryption for data in transit, and controlled access mechanisms. A computing system may further output privacy-preserving metrics (indices, zones, ranges, trend curves, and confidence indicators) without explicit anatomical imagery and without explicit anatomical labeling.

C. On-Device Encryption (Data at Rest) (Non-Limiting)

In various embodiments, the node stores buffered time-series data, session data, metadata, and/or derived metrics in encrypted form using authenticated encryption. Non-limiting approaches include: 1. AES-256-GCM or ChaCha20-Poly1305 for authenticated encryption of stored records; 2. per-record unique nonces/IVs and integrity tags; 3. partitioning storage into encrypted segments including raw sensor data, derived features, and device metadata; and 4. secure erase procedures for deletion of cached data and keys. In certain embodiments, the node encrypts data prior to writing to non-volatile memory such that stored data remains unintelligible without the appropriate key material.

D. Key Management and Secure Storage (Non-Limiting)

In various embodiments, encryption keys are managed using secure key-handling practices, including: 1. generation of device-specific keys at provisioning and/or first initialization; 2. storage of key material in a secure element, trusted execution environment, or protected memory region where available; 3. derivation of session keys using key derivation functions (e.g., HKDF) from a device root key; and 4. key rotation policies and revocation mechanisms for loss scenarios. In certain embodiments, a node does not store long-term plaintext keys in general-purpose storage and may derive transient keys for each session.

E. Secure Communications (Data in Transit) (Non-Limiting)

In various embodiments, communications between a node and a user device or computing system are secured using authenticated encryption and mutual authentication. Non-limiting approaches include: 1. secure pairing and bonding procedures; 2. authenticated key exchange; 3. encryption of transmitted data payloads; and 4. replay protection using nonces, counters, or timestamps. In certain embodiments, communication sessions include integrity checks and device authentication to reduce spoofing and unauthorized access.

F. Access Control, Consent Gating, and Cross-Device Correlation (Non-Limiting)

In various embodiments, access to data and correlation outputs is controlled via explicit consent gating. Non-limiting controls include: 1. user permissioning for data synchronization and sharing; 2. separation of identities for different nodes and optional linking only under explicit consent; 3. limiting correlation outputs to privacy-preserving aggregate metrics rather than raw signals; and 4. additional quality thresholds prior to including any window in cross-device correlation computations.

G. Privacy-Preserving Outputs and Minimization (Non-Limiting)

In various embodiments, user-facing outputs are minimized and abstracted to reduce sensitivity. Outputs may comprise indices, zones, ranges, trend curves, distributions, and confidence indicators rather than raw waveforms. In certain embodiments, sensitive raw data is retained only transiently and purged after feature extraction and/or successful synchronization.

H. Variations

Any embodiment described herein may use different encryption algorithms, key derivation methods, secure elements, and communication protocols, while maintaining the functional intent of protecting data at rest and in transit and enforcing consent-gated access to correlation outputs.